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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 43.7

$$\begin{aligned}
 \int \frac{1-x^4}{3x^5+x^4} dx &= \int \frac{1-x^4}{x^4(3x+1)} dx = \int \frac{1}{x^4(3x+1)} dx - \int \frac{1}{3x+1} dx \\
 \Rightarrow \int \frac{1}{x^4(3x+1)} dx &= \frac{A}{x^4} + \frac{B}{x^3} + \frac{C}{x^2} + \frac{D}{x} + \frac{E}{3x+1} \\
 A &= \left. \frac{1}{3x+1} \right|_{x=0} = 1 \\
 E &= \left. \frac{1}{x^4} \right|_{x=-\frac{1}{3}} = 81 \\
 \frac{1}{x^4(3x+1)} - \frac{1}{x^4} &= \frac{-3}{x^3(3x+1)} \\
 \Rightarrow B &= \left. \frac{-3}{3x+1} \right|_{x=0} = -3 \\
 \frac{-3}{x^3(3x+1)} + \frac{3}{x^3} &= \frac{9}{x^2(3x+1)} \\
 \Rightarrow C &= \left. \frac{9}{3x+1} \right|_{x=0} = 9 \\
 \frac{9}{x^2(3x+1)} - \frac{9}{x^2} &= \frac{-27}{x(3x+1)} \\
 \Rightarrow D &= \left. \frac{-27}{3x+1} \right|_{x=0} = -27 \\
 \Rightarrow \int x^{-4} dx - 3 \int x^{-3} dx + 9 \int x^{-2} dx - 27 \int x^{-1} dx + \frac{80}{3} \ln|3x+1| \\
 &= \frac{1}{-3x^3} + \frac{3}{2x^2} - \frac{9}{x} - 27 \ln|x| + \frac{80}{3} \ln|3x+1| + C
 \end{aligned}$$

References